

MAC Generation and HSM Integration Specification

Purpose: This document explains the existing integration of how the application constructs the MAC request, sends it to the HSM, interprets the HSM response, and places the resulting MAC into ISO8583 field 128.

1. End-to-End Flow

1. Populate all ISO8583 fields that are required to be included in the MAC.
2. Do not generate the MAC before the message is fully populated.
3. Call `data.pack()` to serialize the ISO8583 message into its wire-format representation.
4. Convert the packed bytes to a UTF-8 String.
5. The request is sent to the HSM
6. The HSM returns an M7 response.
7. The application locates M7, reads the HSM error code, and, for error code 00, extracts 8 characters as the MAC.
8. The extracted MAC is assigned to ISO8583 field 128.
9. If the HSM call fails or does not return success, the current implementation places NNNNNNNN in field 128.

2. HSM M6 Request Structure

The request is constructed as the following logical sequence:

Component	Value	Meaning
Message header	00003	Static HSM message header used by this implementation.
Command	M6	MAC generation command.
MAC mode	0	
Input format	0	
MAC algorithm	01	
Padding	1	.
Key indicator	008	
Key scheme	U	Key scheme indicator used before the MAC key.
MAC key	<code>keyEncryptionKey</code>	MAC Key- SCB_ZEK_LMK
Data length	4 hex characters	<code>dataLength</code> converted to hexadecimal and left-padded with zeroes.
MAC input	<code>data</code>	The packed ISO8583 message

3. M7 Response Processing

The response is searched for the literal M7 command marker. If M7 is not found, the method returns the result without a MAC.

When M7 is found, the response is interpreted as follows:

Response Portion	Extraction	Purpose
Before M7	substring(0, headerIndex)	Stored as messageHeader.
M7	substring(headerIndex, headerIndex + 2)	Response code marker.
After M7	substring(headerIndex + 2, headerIndex + 4)	HSM response code.
After response code	substring(headerIndex + 4, headerIndex + 12)	8-character returned MAC when is 00.

4. Success Criteria

The implementation considers the MAC operation successful only when the extracted HSM response code is exactly 00. In that case, the next eight characters are stored as MAC_KEY and returned to the caller.

5. Reference Implementation Summary

In simplified pseudocode, the existing implementation behaves as follows:

```
packed = UTF8(data.pack())
dataLength = length(trim(packed-as-string))

key = SCB_ZEK_LMK

m6Body =
  "00003" +
  "M6" +
  "0" + // single block
  "0" + // binary input
  "01" + // algorithm identifier
  "1" + // zero padding
  "008" + // key indicator
  "U" +
  key +
  HEX4(dataLength) +
  packed

request = Binary2ByteLength(length(m6Body)) + m6Body

response = HSM.send(request)

locate "M7"
errorCode = two characters after M7

if errorCode == "00":
```

mac = next 8 characters
ISO8583.field128 = mac
else:
ISO8583.field128 = "NNNNNNNN"
